

1	180 + 149 or 360 – 31		2	M1
	<i>Working not required, so correct answer scores full marks</i>	329		A1
Total 2 marks				

2	$(BC^2 =) 150^2 + 275^2 - (2 \times 150 \times 275 \times \cos 120) (= 139\,375)$		5	M1 for correct substitution into the cosine rule
	$(BC =) \sqrt{150^2 + 275^2 + 41250}$ or $\sqrt{139375}$ or $25\sqrt{223}$ or 373....			M1 for correct order of operations and square root
	e.g. $\frac{\sin ABC}{275} = \frac{\sin 120}{"373..."}$ or $275^2 = 150^2 + "373..."^2 - (2 \times 150 \times "373..." \times \cos ABC)$ or $\cos ABC = \frac{150^2 + "373..."^2 - 275^2}{2 \times 150 \times "373..."}$ or $\frac{\sin ACB}{150} = \frac{\sin 120}{"373..."}$ or $150^2 = 275^2 + "373..."^2 - (2 \times 275 \times "373..." \times \cos ACB)$ or $\cos ACB = \frac{275^2 + "373..."^2 - 150^2}{2 \times 275 \times "373..."}$			M1 (dep on 1 st M1) ft 373... for a correct trig statement involving angle <i>ABC</i> or angle <i>ACB</i>
	$(ABC =) \sin^{-1} \left(\frac{\sin 120}{"373..."} \times 275 \right) (= 39.6...)$ or $(ABC =) \cos^{-1} \left(\frac{150^2 + "373..."^2 - 275^2}{2 \times 150 \times "373..." } \right) (= 39.6...)$ or $(ACB =) \sin^{-1} \left(\frac{\sin 120}{"373..."} \times 150 \right) (= 20.3...)$ or $(ACB =) \cos^{-1} \left(\frac{275^2 + "373..."^2 - 150^2}{2 \times 275 \times "373..." } \right) (= 20.3...)$			M1 for a complete method to find angle <i>ABC</i> or angle <i>ACB</i>
		140		A1 accept 140 – 140.4
Total 5 marks				

3	(a)	700 ÷ 200 (= 3.5)		3	M1 or 3.5 shown on diagram – within bounds of overlay
					M1 for line drawn at correct angle $\pm 2^\circ$ within bounds of overlay
			<i>C</i> indicated in correct position		A1 for <i>C</i> drawn within bounds of overlay, inclusive of lines.
	(b)		(1 :) 20 000	1	B1
Total 4 marks					